

# The Dependence of Unimolecular Reaction Rates on the Anisotropy of Potential Energy Surfaces

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*Dedicated to Prof. Dr. Heinz Georg Wagner on the occasion of his 60th birthday*

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Simplified statistical adiabatic channel treatments of unimolecular reactions and the reverse recombination or bimolecular capture processes employ the anisotropy ratio  $\alpha/\beta$  between the looseness parameter  $\alpha$  and the standard Morse parameter  $\beta$  of the potential energy surface. The ratio  $2\alpha/\beta$  is now being interpreted as the exponent of a relation between the barrier of the anisotropic potential  $V_o(r)$  and the radial potential  $V(r)$  between the dissociation fragments at distance  $r$ . Comparisons with detailed SACM calculations for various potentials are being made.

In vereinfachten Versionen des statistischen Modells adiabatischer Reaktionskanäle (SACM) für unimolekulare Reaktionen und die entsprechenden Rekombinations- und Einfangprozesse wurde bisher das Anisotropieverhältnis  $\alpha/\beta$  der Potentialfläche verwendet, wobei  $\alpha$  einen „Lockerheitsparameter“ und  $\beta$  den Standard-Morseparameter darstellt. Das Verhältnis  $2\alpha/\beta$  wird jetzt als Exponent der Beziehung zwischen der Barriere  $V_o(r)$  des Anisotropieanteils und dem Radialanteil  $V(r)$  des Potentials zwischen den Dissoziationsprodukten im Abstand  $r$  interpretiert. Vergleiche mit detaillierten SACM-Rechnungen für verschiedene Potentiale werden vorgenommen.

## 1. Introduction

Unimolecular bond fission reactions and the reverse bimolecular recombination or capture processes are markedly influenced by anisotropies of the potential energy surfaces on which they take place. If the potentials were isotropic, phase space theory (PST) or transition state theory (TST) in its loose limit would apply. An analysis of experimental results on thermal unimolecular bond fissions and the reverse recombinations in the high pressure limit [1] has shown, however, that PST always overestimates the

rates. Representing the rate coefficients  $k$  by the isotropic expression  $k^{\text{PST}}$  from PST and a "rigidity factor"  $f^{\text{rigid}}$ ,

$$k = f^{\text{rigid}} k^{\text{PST}}, \quad (1)$$

values of  $f^{\text{rigid}}$  below unity were found for all neutral reaction systems. Ion-molecule capture processes are showing an analogous behaviour. Only for the purely isotropic charge-polarizability potential the capture rate coefficient is given by the Langevin expression corresponding to PST. The presence of a permanent dipole moment with its anisotropic potential reduces the rate coefficient markedly below the "locked dipole" expression corresponding to PST [2].

Obviously one would like to have a simple way to express the rigidity factors  $f^{\text{rigid}}$  from Eq. (1) in terms of the anisotropy properties of the potential such as the barriers  $V_o(r)$  of the anisotropic part of the potential at the distance  $r$  between the reactants B and C in a  $A \rightleftharpoons B + C$  reaction. Alternatively, the anisotropy of the potential might be characterized by the  $r$ -dependent frequencies  $\nu(r)$  of the transitional modes of A becoming product rotors during the reaction. On the other hand, the isotropic PST expression  $k^{\text{PST}}$  of the rate coefficient in Eq. (1), being governed exclusively by the radial part  $V(r)$  of the potential, has also to be elaborated in detail.

The determination of the rigidity factor  $f^{\text{rigid}}$ , of course, implies the complete solution of the kinetic problem. Therefore, one may try to approach the task on various levels of approximation. In the following we do this in the framework of the statistical adiabatic channel model [3] (SACM) which may be understood as a "state-resolved" or "channel-resolved" version of TST. The concept of this approach is easily formulated: By fixing  $r$ , the eigenvalues  $V_i(r)$  of the system are determined, defining "adiabatic channel potential curves". The maxima of these curves along  $r$  identify the "channel threshold energies"  $E_{oi}$ . At an energy  $E$ , channels with  $E_{oi} \leq E$  are "open". Assuming equal statistical weight for all open channels, i.e. using the general postulate of statistical rate theories, then all rate parameters are expressed in the usual TST formalism.

Adiabatic channel potential curves  $V_i(r)$  for isotropic potentials, such as used in the PST limit, are given by

$$V_i(r) = V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} + E_i(\infty) \quad (2)$$

where  $V(r)$  denotes the radial potential,  $E_i(\infty)$  are the rovibrational eigenvalues of the separated fragments  $B + C$ ,  $\mu$  is the reduced mass of  $B + C$ , and  $l$  is the quantum number of orbital rotation between B and C. The maxima of  $V_i(r)$  from Eq. (2) are expressed by

$$E_{oi}^{\text{PST}} = E_i(\infty) + E_o(l) \quad (3)$$

where  $E_o(l)$  are the centrifugal barriers. Anisotropy of the potential energy surface raises  $E_{oi}$  to values higher than  $E_{oi}^{\text{PST}}$ .

In order to avoid the cumbersome accurate eigenvalue solution for all  $V_i(r)$ , in ref. [3] an exponential eigenvalue interpolation scheme was applied,

$$V_i(r) \approx V(r) + E_i(r_e) \exp(-\alpha(r - r_e)) + E_i(\infty)[1 - \exp(-\alpha(r - r_e))] + B(r)P(P + 1) \quad (4)$$

with an interpolated rotational quantum number

$$P = J \exp(-\alpha(r - r_e)) + l[1 - \exp(-\alpha(r - r_e))]. \quad (5)$$

Here,  $r_e$  denotes the equilibrium distance of  $r$  in the reactant A,  $E_i(r_e)$  are the reactant rovibrational eigenvalues [excluding the reaction coordinate and the rotation with an axis perpendicular to it (with quantum number  $J$ )], and  $\alpha$  defines a “looseness parameter” of the potential. The exponential switching model [4] goes back to BEBO-type Pauling rules [5]. A closer inspection of the simple SACM using Eqs. (4) and (5) in refs. [6–8] showed that, with a Morse potential

$$V(r) = -2D \exp(-\beta(r - r_e)) + D \exp(-2\beta(r - r_e)), \quad (6)$$

the results of the treatment could be represented in a relatively simple way as a function of the reactant A and fragment B + C properties, and the two parameters  $\alpha$  and  $\beta$ . The “anisotropy ratio”  $\alpha/\beta$ , in particular, allowed to express the deviations from PST. For  $\alpha/\beta \gtrsim 1$ , PST was approached; for  $\alpha/\beta \approx 0$ , rigid TST was reproduced.

By separating centrifugal effects, i.e. evaluating  $E_{oi}$  for  $J = 0$  with Eq. (4) and inserting the results for  $E_{oi}$  into Eq. (2) instead of  $E_i(\infty)$ , a particularly simple version of SACM was derived in refs. [6–8]. For separated transition modes, in this way numbers of open channels  $W(E, J)$ , i.e. numbers of channels with  $E_{oi} < E$ , were expressed by

$$W(E, J) \approx \left\{ \frac{E - E_o(J)}{\varepsilon(\alpha/\beta)} \right\}^{x(\alpha/\beta)} \quad (7)$$

where the exponents  $x$  and the “quanta”  $\varepsilon$  parametrically depend on  $\alpha/\beta$  and the reactant and product properties such as the Morse dissociation energy  $D$ , the quantum  $\varepsilon_e$  of the transitional mode at  $r = r_e$  and the fragment rotational constant  $B_\infty$  at  $r \rightarrow \infty$ . The dependencies of  $\varepsilon$  and  $x$  on  $\alpha/\beta$  were obtained by fitting the numerical results of the  $E_{oi}$  determination in the form of

$$\varepsilon \approx B_\infty + (\varepsilon_e - B_\infty) \exp[-C_e \alpha/\beta - C_4(\alpha/\beta)^4] \quad (8)$$

$$C_3 \approx 4.00 - 1.29 \ln \varepsilon_e/D$$

$$C_4 \approx 2.80 - 5.19 \ln \varepsilon_e/D$$

$$x \approx x_e \exp(-y - y^n) + x_\infty [1 - \exp(-y - y^n)] \quad (9)$$

$$y \approx C_2 \alpha / \beta$$

$$C_2 \approx 1.24 + 55 B_\infty / \varepsilon_e$$

$$n \approx 2.25 + 0.5 \times 10^{-3} \varepsilon_e / B_\infty.$$

For a transition mode, which corresponds to an oscillator in the reactant A and to a rotor in the separated fragments, one has  $x_e = 1$  and  $x_\infty = 1/2$ . With Eq. (7), activated complex partition functions are directly obtained [6] in the form

$$Q^* \approx \Gamma(1+x)[1 - \exp(-\varepsilon/kT)]^{-x} \approx \Gamma(1+x)[kT/\varepsilon]^x \quad (10)$$

where the latter approximation is valid for  $kT \gg \varepsilon$ . By separation of coordinates in the described way, a number of couplings are violated which in refs. [6–8] were corrected for a posteriori in “a global way”. A comparison with an SACM treatment, avoiding this separation but still using the exponential eigenvalue scheme of Eqs. (4) and (5), was given for the  $\text{H}_2\text{O}_2 \rightleftharpoons 2\text{OH}$  system in ref. [8]. There were differences in the results; however, these were small and could be removed by minor adjustments of the fit parameter  $\alpha/\beta$ .

As long as no unique relation between the real potential and the “global parameters”  $\alpha$  and  $\beta$  is established, the described SACM treatment remains on the “two-parameter fit” level of unimolecular rate theory [9]. In this way one generally very well correlates experimental results, even when unphysical models for activated complexes are chosen. However, when one knows more about the potential, of course, one would like to implement this information as well. In particular, one would like to choose  $\alpha$  and  $\beta$  values corresponding to a given potential. It is the aim of the present article to provide guidelines for this procedure.

## 2. Simplified interpretation of the anisotropy ratio $\alpha/\beta$

In recent studies [9–12], for selected systems, we have calculated explicit adiabatic channel potential curves on ab initio potential energy surfaces avoiding the simplifications made in earlier SACM treatments. Analogous calculations have also been performed in refs. [13–17]. These accurate adiabatic channel calculations now may serve to control the extent of approximation introduced by the earlier simplified SACM treatments. Before such comparisons are made, we generalize the earlier SACM treatment trying to assign a more general meaning to the anisotropy ratio  $\alpha/\beta$ .

We start on the level of the eigenvalue interpolation scheme such as given by Eqs. (4) and (5). However, we replace the exponential switching term  $\exp(-\alpha(r - r_e))$  by a general relation between the radial potential

$V(r)$  and the  $r$ -dependent oscillator quantum  $\varepsilon(r)$  for a transition mode going from a harmonic oscillator at  $r \approx r_e$  to a product rotor at  $r \rightarrow \infty$ . Likewise, one may express the harmonic oscillator quantum  $\varepsilon(r)$  by the barrier  $V_o(r)$  of the anisotropic potential using the relation [18]

$$\varepsilon(r) \approx (V_o(r)N^2B)^{1/2} \quad (11)$$

which is valid for an angular potential of the  $\cos N\theta$  type, i.e. for

$$V(\theta) = V_o(r)[1 - \cos N\theta]/2 \quad (12)$$

and an effective rotational constant  $B$ .

We express  $\varepsilon(r)$  by the relation

$$\varepsilon(r) \approx \varepsilon_e (|V(r)|/2D)^{\alpha/\beta} \quad (13)$$

which includes the relation between the long-range term of a Morse potential  $V(r) \approx -2D \exp[-\beta(r - r_e)]$  and an exponential eigenvalue switching such as used in Eq. (4). For  $r \rightarrow r_e$ , Eq. (13) would give  $\varepsilon = 0.707 \varepsilon_e$ ; however, the long-range part of the potential governs most of the relevant threshold energies. Eq. (13) via Eq. (11) is equivalent to

$$V_o(r) \approx \varepsilon_e^2 (|V(r)|/2D)^{2\alpha/\beta} / B_\infty \quad (14)$$

(for  $N = 1$  and  $B \approx B_\infty$ ). Employing Eqs. (13) and (14), we define the effective anisotropy ratio  $2\alpha/\beta$  of the potential surface via the exponent of the relation

$$V_o(r) \propto |V(r)|^{2\alpha/\beta} \quad (15)$$

between the anisotropy barrier  $V_o(r)$  and the radial potential  $|V(r)|$ . However, besides the exponent also the other factors in Eqs. (13) and (14), which are given by reactant and product properties, are of quantitative relevance.

At first we consider the case  $J = 0$  and we neglect the centrifugal term, i.e. we put  $P = 0$  in Eq. (4). Instead of Eq. (4), from Eqs. (13) and (14) we now obtain

$$V_i(r) \approx -|V(r)| + (i + 1/2)\varepsilon_e (|V(r)|/2D)^{\alpha/\beta} + i(i + 1)B_\infty [1 - (|V(r)|/2D)^{\alpha/\beta}]. \quad (16)$$

For  $i \gg 1$ , Eq. (16) is approximated by

$$V_i(r) - E_i(\infty) \approx -|V(r)| + \left\{ \frac{i\varepsilon_e - i^2 B_\infty}{(2D)^{\alpha/\beta}} \right\} |V(r)|^{\alpha/\beta}. \quad (17)$$

The derivation of  $V_i(r)$  as a function of  $|V(r)|$ , without further need to specify details of the radial potential, after few algebraic steps leads to the adiabatic channel threshold energies

$$E_{oi} - E_i(\infty) \approx \left[ \left( \frac{\alpha}{\beta} \right)^{-1} - 1 \right] \left( \frac{\alpha}{\beta} \right)^{\frac{1}{1-\alpha/\beta}} \left\{ \frac{i\varepsilon_e - i^2 B_\infty}{(2D)^{\alpha/\beta}} \right\}^{\frac{1}{1-\alpha/\beta}}. \quad (18)$$

The channel threshold energies are located at a distance  $r$  where

$$V(r) \Big|_{E_{oi}} \approx - \left\{ \left( \frac{\alpha}{\beta} \right) \left[ \frac{i\epsilon_e - i^2 B_\infty}{(2D)^{\alpha/\beta}} \right] \right\}^{\frac{1}{1-\alpha/\beta}}. \quad (19)$$

For not too high channel numbers  $i$  (where centrifugal effects also would have to be included, even [6] for  $J = 0$ ), one may further neglect  $i^2 B_\infty \ll i\epsilon_e$  such that the channel threshold energies are found at a distance  $r$  where

$$\epsilon(r) \Big|_{E_{oi}} \approx \frac{|V(r)|}{i\alpha/\beta} \quad (20)$$

or

$$V_o(r) \Big|_{E_{oi}} \approx \left\{ \frac{|V(r)|}{i\alpha/\beta} \right\}_{B_\infty}^2. \quad (21)$$

Eqs. (19)–(21) will be helpful in identifying the potential range which is relevant for the channel threshold energies.

Eq. (18) provides a direct access to the number of open channels  $W(E, J = 0)$ , since  $i$  (for  $i \gg 1$ ) is approximately equal to  $W$  and  $E_{oi}$  defines the energy scale  $E$ . Resolving Eq. (18) with respect to  $W(E, J = 0)$ , for  $i\epsilon_e \gg i^2 B_\infty$ ,  $E_i(\infty) \approx i^2 B_\infty$  and for  $\alpha/\beta = 0.5$ , leads to

$$W(E, J = 0) \approx \left\{ \frac{E - E_o(J = 0)}{B_\infty + \epsilon_e^2/8D} \right\}^{1/2}. \quad (22)$$

For  $\alpha/\beta \rightarrow 0$ , one obtains the oscillator result  $W(E, J = 0) \approx [E - E_o(J = 0)]/\epsilon_e$ ; for  $\alpha/\beta \rightarrow \infty$ , the PST-rotor limit  $W(E, J = 0) \approx \{[E - E_o(J = 0)]/B_\infty\}^{1/2}$  is approached. For intermediate values of  $\alpha/\beta$ , in the ranges  $0 < \alpha/\beta < 0.5$  and  $0.5 < \alpha/\beta < 1$ , the energy dependences of  $W(E, J = 0)$  take a more complex form which differs from Eq. (7). Eqs. (7)–(9) neglected this complex energy dependence by using only the range  $10 \lesssim W(E, J = 0) \lesssim 20$ . For this range, a comparison with the present analytical approach shows good agreement as well. For the most important range of values of  $\alpha/\beta$  near 0.5, Eqs. (7)–(9) give

$$\epsilon(\alpha/\beta = 0.5) \approx B_\infty + \epsilon_e^2/8.8D \quad (23)$$

which again agrees well with the present result of Eq. (22). The small discrepancy is due to the account for centrifugal effects in the derivation of Eqs. (7)–(9) [with  $l = i$  in Eq. (5)] which results in a minor reduction of  $W(E, J = 0)$  as compared to Eq. (22).

Eqs. (18) and (19) clearly show that, with increasing channel quantum number  $i$ , the threshold energies increase and their position moves inward (i.e. toward smaller  $r$ ). A comparison of the vibrational and the rotational terms in Eq. (16) also shows that, with increasing  $i$ , the vibrational character of the channel potential curves increases. Whether a purely vibrational

model is sufficient for all channels, i.e. whether the third term at the rhs of Eq. (16) can be neglected, depends on the value of the ratio  $\varepsilon_e^2/4B_\infty D$ . If the ratio is much larger than unity, this is case; otherwise, the hindered rotor character of the channel potential curves dominates. The more detailed explicit treatment of channel potential curves in the next section shows that, in general, oscillator- and rotor-type potential curves contribute simultaneously.

The treatment so far has led to an energy pattern of the channel threshold energies  $E_{oi}$  given by Eq. (18). For  $\alpha/\beta \approx 0.5$ , the simple expression (22) for the number of open channels  $W(E, J = 0)$  then follows from Eq. (18). It corresponds to a rotor-type activated complex model in which the effective rotational constant is given by

$$\varepsilon(\alpha/\beta = 0.5) \approx B_\infty + \varepsilon_e^2/8D \quad (24)$$

or by the result of Eq. (23) from the numerical analysis accounting for some centrifugal contributions. The given treatment has neglected two important classes of couplings, i.e. the couplings between several transition modes, and the coupling between the transition modes and overall rotation for  $J \neq 0$ . We shall discuss later on the consequences of neglecting these couplings and the corrections required. Before we do this and before we consider explicit adiabatic channel calculations, we shall briefly summarize the results obtained so far:

(i) The anisotropy ratio  $2\alpha/\beta$  of the SACM version of refs. [1, 3, 6–8] can be interpreted as the exponent of the anisotropy barrier  $V_o(r) \Leftrightarrow$  radial potential  $V(r)$  relationship of Eq. (15).

(ii) The derivation of the rigidity factors  $f^{\text{rigid}}$  in Eq. (1), or the number of open channels  $W(E, J)$ , on this level is practically independent of the type of the radial potential [i.e. its radial dependence given by the exponent in a  $V(r) \propto r^{-n}$  representation]. However, the coefficients in  $V_o(r) \propto |V(r)|^{2\alpha/\beta}$  or  $\varepsilon(r) \propto |V(r)|^{\alpha/\beta}$  relationships such as Eqs. (13) and (14) are quantitatively entering  $f^{\text{rigid}}$  and  $W(E, J)$ .

(iii) The SACM treatment of refs. [6–8] and the analysis of experimental data in ref. [1] are characterized by a number of simplifications, whose consequences in part compensate each other: The use of a centrifugal partition function entering  $k^{\text{PST}}$  on the basis of a Morse radial potential neglects the specific long-range forces.

The decoupling of centrifugal effects and the decoupling of various transition modes is of major importance. Global empirical corrections for these errors have been discussed in refs. [6–8]. To the extent that the corresponding errors and the errors in the centrifugal partition function do not compensate each other, they are in practice compensated by an adjustment of the ratio  $\alpha/\beta$ .

(iv) The quasi-universal result  $2\alpha/\beta \approx 1$  from the analysis in ref. [1] would suggest that the described artifacts are all not of central importance.

A value of  $2\alpha/\beta = 1$  characterizes long-range potentials with one dominant term in the  $r^{-n}$ -expansion [19]. Further discussions of the role of the potential are given in the following section.

(v) The coefficients of the  $V_o(r) \langle = \rangle V(r)$  relation are of quantitative importance. Eqs. (13) and (14), therefore, have to be controlled with ab initio potentials. Specific effects of individual potential not only enter via the centrifugal partition function based on the radial potential, but also via the exponent  $2\alpha/\beta$  and the coefficients of  $V_o(r) \langle = \rangle V(r)$  relations.

### 3. Accurate adiabatic channel potential curves. Results for small $j$

In the following we proceed to an explicit channel eigenvalue calculation replacing the interpolating approximation of Eqs. (4) and (16). It was shown in ref. [18] by RKR-inversion that the eigenvalue pattern of Eq. (4) corresponds to a truncated tg-type angular potential. This is not very realistic. Instead, we now consider cos-type angular potentials. Eigenvalues  $E_i(r)$  for a

$$V(\theta) = V_o(r) (\cos \theta)/2 \quad (25)$$

angular potential have been elaborated in the discussion of the Stark effect of dipole molecules [20–24]. Tabulations of  $E_i(r)$  have been given in the form of  $\varepsilon(\lambda)$  for  $j \leq 4$  in refs. [20] and [22] where  $\lambda$  corresponds to  $V_o(r)/2B_\infty$  and  $\varepsilon$  to  $E_i/B_\infty - V_o(r)/2B_\infty$ . Beyond the tabulated values asymptotic series expansions are useful: for small values of  $\lambda$ , perturbation theory gives

$$\frac{E_i}{B_\infty} \approx j(j+1) + \frac{1}{2} \frac{V_o}{B_\infty} - \frac{1}{8} \left( \frac{V_o}{B_\infty} \right)^2 F(j, m) + \frac{1}{16} \left( \frac{V_o}{B_\infty} \right)^4 K(j, m) \quad (26)$$

for a linear fragment interacting with an atom. The quantities  $F(j, m)$  and  $K(j, m)$ , with  $|m| \leq j$ , are given by

$$F(j, m) = \frac{3m^2 - j(j+1)}{j(j+1)(2j-1)(2j+3)} \quad (27)$$

and

$$\begin{aligned} 8K(j, m) = & \frac{[(j-1)^2 - m^2][j^2 - m^2]}{(2j-3)(2j+1)j^2(2j-1)^3} \\ & - \frac{[(j+1)^2 - m^2][(j+2)^2 - m^2]}{(2j+1)(2j+5)(j+1)^2(2j+3)^3} \\ & + \frac{[j^2 - m^2][(j+1)^2 - m^2]}{(2j-1)(2j+3)(2j+1)^2j^2(j+1)^2} \\ & + \frac{[(j+1)^2 - m^2]}{(2j+3)^2(2j+1)^2(j+1)^3} - \frac{[j^2 - m^2]}{(2j+1)^2(2j-1)^2j^3}. \end{aligned} \quad (28)$$

For large values of  $\lambda$ , an anharmonic oscillator expansion gives [25]

$$\frac{E_i}{B_\infty} \approx (2j - |m| + 1) \sqrt{\frac{V_o}{B_\infty}} + \frac{b(j, m)}{4} + \frac{c(j, m)}{32 \sqrt{V_o/B_\infty}} \quad (29)$$

with

$$b(j, m) = m^2 + 2|m|j - 2j^2 + |m| - 2j - 2 \quad (30)$$

and

$$c(j, m) = -4|m|^3 + 6m^2j + 6|m|j^2 - 4j^3 + 3m^2 + 6|m|j - 6j^2 + 3|m| - 6j - 2. \quad (31)$$

A symmetric top fragment interacting with an atom has also been treated explicitly by second order perturbation theory. Here,  $j(j+1)$  in Eq. (26) is replaced by  $j(j+1) + (A_\infty/B_\infty - 1)k^2$  with  $|k| \leq j$  and  $F(j, m)$  is replaced by  $F(j, m, k)$  with

$$F(j, m, k) = - \left[ \frac{(j^2 - k^2)(j^2 - m^2)}{j^3(2j-1)(2j+1)} - \frac{[(j+1)^2 - k^2][(j+1)^2 - m^2]}{(j+1)^3(2j+1)(2j+3)} \right] \quad (32)$$

( $A_\infty$  and  $B_\infty$  are the rotational constants of the symmetric top). The anharmonic oscillator expansion is modified [22, 25] by replacing  $|m|$  in the first term of Eq. (29) by  $|m+k|$ . A perturbation treatment is also available for two interacting linear fragments [19].

Eigenvalues  $E_i(r)$  for  $\cos^2$ -type angular potentials follow from the solution of the spheroidal wave equation [26–29]. Tabulations for small  $j$  and  $m$  as well as asymptotic series expansions were given. In the present article, for illustration, we shall limit our attention to the  $\cos \theta$ -potential.

Working in a body-fixed coordinate system and applying the centrifugal sudden approximation [30], the channel quantum number  $m$  is associated with the projection of both the total angular momentum (quantum number  $J$ ) and the fragment rotational angular momentum (quantum number  $j$ ) on the line connecting the centers of mass of the two fragments. On the same time, angular momentum conservation requires

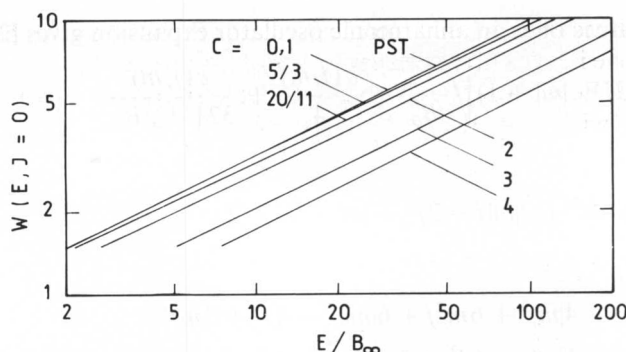
$$\vec{l} + \vec{j} = \vec{J} \quad (33)$$

such that

$$|J - j| \leq l \leq J + j \quad (34)$$

with the orbital angular momentum quantum number  $l$ .

We shall consider two different situations, one in which small total  $l$  are involved such that centrifugal contributions are nearly negligible and, in



**Fig. 1.** Numbers of open channels  $W(E, J = 0)$  as a function of the potential parameter  $C$  for an (atom + linear fragment)-system [see Eq. (36) with  $2\alpha/\beta = 1$ ;  $W(E, J)$  has been “smoothed”, i.e. the step from  $W$  to  $W + N$  is drawn at  $W + N/2$ ; the curves for  $C = 0$  and 1 correspond to phase space theory, i.e. potentials without anisotropy].

the next section, one where large values of  $J$  and  $l$  also become relevant. For the first case, the channel potentials are given by

$$V_i(r) \approx V(r) + E_i(r) \quad (35)$$

with  $E_i(r)$  from Eqs. (26)–(31). Relating the angular barrier  $V_o(r)$  to the radial potential  $|V(r)|$  by analogy to Eq. (14), i.e. employing

$$\frac{V_o(r)}{B_\infty} \approx C \left( \frac{|V(r)|}{B_\infty} \right)^{2\alpha/\beta} \quad (36)$$

the derivation of  $V_i(r)$  with respect to  $|V(r)|$  leads to the channel threshold energies independent of the particular shape of  $|V(r)|$  [as long as  $V(r)$  is a monotonic function of  $r$ ]. Counting the corresponding numbers of open channels, one obtains results such as shown in Fig. 1 for  $J = 0$ ,  $\alpha/\beta = 0.5$ , and various values of  $C$ .

The curves in Fig. 1 for  $J = 0$ , according to the meaning of the quantum number  $m$  (see above), all correspond to  $m = 0$ . A number of observations can be made:

- (i) For  $C \leq 1$ , the phase theoretical result, i.e.  $C = 0$ , is obtained.
- (ii) For  $1 < C < 2$ , the  $W(E, J)$  curves at small energies follow the PST results for a while, before they fall below the PST curves approaching a curve corresponding to PST but with a larger effective rotational constant, i.e.  $B_{\text{eff}} > B_\infty$ .
- (iii) The transition from the low-energy curves (PST with  $B_{\text{eff}} = B_\infty$ ) and the high energy curves (PST with  $B_{\text{eff}} > B_\infty$ ) occurs only in the range  $1 < C < 2$ . For  $C \geq 2$ , the high-energy curves are reached already at  $W \leq 2$ . For  $C > 2$ , one also notes adiabatic zero-point barriers, i.e.  $E_{oi} > 0$  for  $W = 1$ .

(iv) With increasing  $C$  and increasing energy the positions of the channel maxima move inward (i.e. to larger values of  $|V(r)|$ ) such that the oscillator expansion of Eq. (29) applies. Using only the harmonic oscillator term and identifying  $j$  with  $W + 1$ , one would obtain

$$W(E, J = 0) \approx \sqrt{\frac{E}{C B_{\infty}}} \quad (37)$$

in comparison to the PST result

$$W(E, J = 0) \approx \sqrt{\frac{E}{B_{\infty}}} \quad (38)$$

Retaining all three terms in the anharmonic oscillator expansion of Eq. (29), such as in the calculations for  $j > 4$  contributing to Fig. 1, the high-energy expression slightly differs from Eq. (37). Instead one has

$$W(E, J = 0) \approx \sqrt{\frac{E}{B_{\text{eff}}}} \quad (39)$$

with

$$B_{\text{eff}}/B_{\infty} \approx \text{Max} \{1, C - 1/2\}. \quad (40)$$

Obviously an SACM-modified PST [8] with  $B_{\text{eff}} > B_{\infty}$  applies again quite well. However, near to the threshold energy, there is a deviation for the range of parameters  $1 < C < 2$  where  $W(E, J = 0)$  at low energies follows the PST result.

Such an effect (much more pronounced because of several transition modes) has recently been observed experimentally [31] in measurements of the dissociation rate constants  $k(E, J = 0) = W(E, J = 0)/h\rho(E, J = 0)$  of jet-cooled NCNO molecules (see e.g. Fig. 9 in ref. [9]). The present SACM treatment provides a plausible explanation of this effect in terms of a statistical theory. The fine details of this effect are governed by the anisotropy of the potential which is here expressed by the parameter  $C$ .

The results from Fig. 1 and Eqs. (37)–(40) do not well correspond to Eqs. (7) and (22) derived from simplified SACM versions although an increasing decrease of  $W(E, J = 0)$  below the PST result with increasing energy was also recognized in ref. [6]. Taking into account that the parameter  $C$  can be related to  $\varepsilon_e$ ,  $B_{\infty}$ , and  $D$  via

$$C = \frac{(\varepsilon_e/D)^2}{2B_{\infty}/D}, \quad (41)$$

such that Eq. (24) corresponds to

$$\varepsilon(\alpha/\beta = 0.5) \approx B_{\infty} (1 + C/4), \quad (42)$$

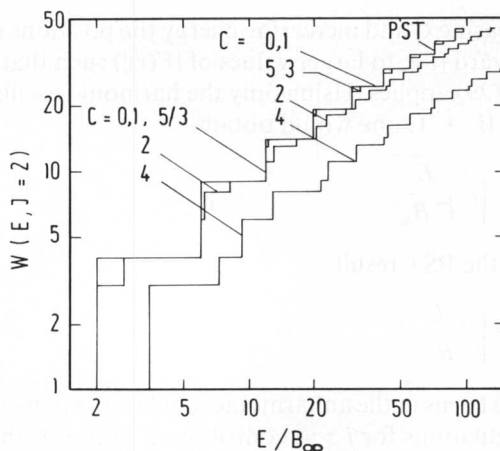


Fig. 2.  $W(E, J = 2)$  [for representation see Fig. 1;  $W(E, J)$  has not been smoothed here].

one notices a markedly larger effective  $B_{\text{eff}} = \varepsilon$  in Eq. (40) than given by the result (42) from the simplified SACM. This difference is due to the “selection” of only  $m = 0$  channels by the  $J = 0$  constraint. The  $m = 0$  channels correspond to the most rigid activated complexes within the  $2j + 1$  channels for a given  $j$ . Only when larger  $|m|$  are included, i.e. at  $J > 0$ , the average activated complex becomes less rigid such that Eq. (42) is approached. The channels with  $|m| = j$  are loosest; Eq. (42) corresponds to an average rigidity of the set of  $m$ -channels of a given  $j$ .

In order to illustrate the decreasing effective rigidity when larger  $|m|$  contribute, Figs. 2 and 3 show results computed for  $W(E, J = 2)$  and  $W(E, J = 4)$  (again neglecting centrifugal energies). Figs. 2 and 3 again include the result from phase space theory for atom + linear molecular fragments (see Eqs. (C14) and (C15) from ref. [7]), i.e.

$$W(E, J) = (2J + 1)(j_{\text{max}} + 1) - J(J + 1) \quad (43)$$

for  $0 \leq J \leq j_{\text{max}}$  and

$$W(E, J) = (j_{\text{max}} + 1)^2 \quad (44)$$

for  $J \geq j_{\text{max}}$  where  $j_{\text{max}}$  is the largest  $j$  with  $j(j + 1) \leq E/B_{\infty}$ . From Figs. 2 and 3 again a number of observations can be made:

(i) For  $C < 1$ , the PST result is obtained. For  $1 < C < 2$ , the curves start on the PST curve close to the threshold before they fall below approaching a PST-type curve with  $B_{\text{eff}} > B_{\infty}$ .

(ii) For  $J > 0$ , channels with  $|m| \leq J$  contribute. As long as the energies are lower than  $B_{\infty}J(J + 1)$ , the results can well be represented with a  $B_{\text{eff}}$  given by Eq. (42). I.e., as long as no  $m$ -channels are excluded by the  $|m|$

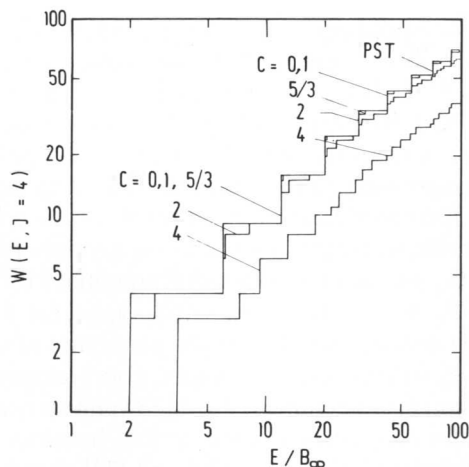


Fig. 3.  $W(E, J = 4)$  [for representation see Fig. 1;  $W(E, J)$  has not been smoothed here].

$\leq J$  constraint, the simplified SACM result gives an excellent average representation of the channel threshold energies.

(iii) For  $E > B_\infty J(J + 1)$ , an increasing number of the looser  $m$ -channels with  $|m| > J$  are excluded with increasing  $j$  such that the result corresponds to more rigid activated complexes. Then  $B_{\text{eff}}$  increases with energy until the high energy limit characterized by Eqs. (39) and (40) again is obtained. Figs. 2 and 3 include the limiting curves of Eqs. (39), (40), and (42).

The presented discussion shows that simplified SACM versions approach the results from detailed channel calculations in many aspects. However, several fine details, especially near to the threshold energy (if  $C < 2$ ) and at  $E > B_\infty J(J + 1)$  are not well reproduced such that a more detailed treatment, such as presented here, is required.

#### 4. Accurate adiabatic channel potential curves. Results for large $J$

So far we have neglected centrifugal effects. Dissociation lifetimes measurements in jet-cooled molecules may correspond to this case such as elaborated in section 3. If one considers thermal dissociation and recombination reactions, however, often large  $l$  values contribute and centrifugal effects have to be taken into account. Now, Eq. (35) has to be replaced by

$$V_i(r) \approx V(r) + E_i(r) + l(l + 1)\hbar^2/2\mu r^2. \quad (45)$$

Eq. (45) assumes that the transition of the centrifugal energy to the limiting long-range expression, see Eqs. (4) and (5), remains adequate. (At very large  $J$  and  $E$ , this may not be the case.) One now has to determine the

dependence of the threshold energies  $E_{oi}(l)$  on  $l$  where  $i$  stands for product quantum numbers such as  $j$  and  $m$ . For convenience we shall neglect vibrational excitation of the fragments such as this also was done in section 3.

Examples for the  $l$ -dependence of the threshold energies  $E_{oi}(l)$  were given and illustrated in refs. [9] and [10] for the charge-dipole potential, i. e. for a potential corresponding to  $C = 2$ . In the following we consider more general potentials. The numerical treatment for the atom + linear fragment case presents no problems when the channel eigenvalues from Eqs. (26)–(31) can be used in Eq. (45) with known radial potential  $V(r)$  and anisotropy barrier  $V_o(r)$ . For  $V_o/B \rightarrow 0$ , Eqs. (26)–(28) apply; for  $V_o/B \rightarrow \infty$ , one uses Eqs. (29)–(31) instead. Apart from the possibility of using numerical accurate values of  $E_i$  in between these limits, also a change from the one to the other expression may be employed at  $V_o/B$  values where the difference between both results (up to the second terms) becomes minimum. An inspection of the tabulated  $E_i$  in refs. [20] and [22] shows that this is an often acceptable simplification. For illustration, in the following we again treat simple cases in which  $V(r)$  and  $V_o(r)$  are connected by Eq. (36) with  $\alpha/\beta = 0.5$  and  $V(r) \propto r^{-n}$ . Then, the centrifugal energy  $E_{cent}(r)$  can be expressed in a simple way by the radial potential  $V(r)$ ,

$$\frac{E_{cent}(r)}{B_\infty} = \frac{l(l+1)\hbar^2}{2\mu B_\infty r^2} = l(l+1)C^* \left( \frac{|V(r)|}{B_\infty} \right)^\gamma = G(l) \left( \frac{|V(r)|}{B_\infty} \right)^\gamma, \quad (46)$$

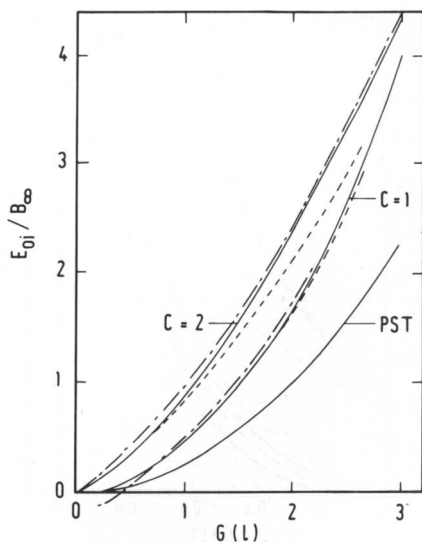
where we have used the abbreviation

$$G(l) = l(l+1)C^*. \quad (47)$$

Channel threshold energies  $E_{oi}$  as a function of  $G(l)$  were illustrated in ref. [9] for the cases  $\gamma = 1$  and  $C = 2$ .

It was shown that “repulsive channels” with  $F(j, m) \leq 0$ , for all  $G(l)$ -values could well be represented by the anharmonic oscillator expansion up to the  $b(j, m)/4$ -term. The “attractive channels” with  $F(j, m) > 0$  could be approximated by this expansion only for  $G(l) > 0.5$ . For  $G(l) < 0.5$ , the perturbation treatment up to the  $F(j, m)\lambda^2/2$ -term provided a more accurate approximation. For low temperature applications this approximation is most relevant. In the following we demonstrate the effect of various potential parameters.

At first, Fig. 4 shows the dependence of  $E_{oi}$  on  $G(l)$  for the ( $j = 0; m = 0$ ) channel with different  $C$  values and  $\gamma = 1$ . Quite analogous to the observations of the last section on  $W(E, J)$ , the case  $C = 1$  approaches the results from phase space theory where  $E_{oi} = j(j+1)B_\infty$  for  $\gamma = 1$ , as long as  $G < 1$ , and  $E_{oi} \rightarrow \infty$  if  $G \geq 1$ . For  $C = 5/3$ , the phase space result is obtained only as long as  $G(l) \leq 1/6$ . For larger  $G$ ,  $E_{oi}$  at first follows the perturbation result before, near  $G \approx 0.6$ , it switches to the oscillator result.



**Fig. 4.** Dependence of the threshold energies  $E_{oi}$  for the ( $j = 0, m = 0$ ) channel of an (atom + linear fragment) system on the orbital angular momentum  $l$  [for  $G(l)$ , see Eqs. (46) and (47);  $\gamma = 1$ , i.e.  $V(r) \propto r^{-2}$  and  $E_{cent} \propto r^{-2}$ ; for potential parameter  $C$ , see Eq. (36) with  $2\alpha/\gamma = 1$ . —: accurate results, ---: oscillator expansion of Eq. (49), - - -: perturbation expansion of Eq. (48), PST results identical with  $C = 1$  and 0].

For  $C = 4$ , there is an adiabatic zeropoint barrier, i.e.  $E_{oi} > 0$  at  $G(l) = 0$ . At small  $G(l)$ -values,  $E_{oi}$  here is also given by the oscillator result. For  $\gamma = 1$ , the perturbation expansion leads to

$$\frac{E_{oi}}{B_{\infty}} \approx j(j+1) + \frac{(1 - 2/C + 2G(l)/C)^2}{2F(j, m)} \quad (48)$$

[and  $E_{oi}/B_{\infty} = j(j+1)$  as long as  $G(l) < (1 - C/2)$ ], the oscillator expansion gives

$$\frac{E_{oi}}{B_{\infty}} \approx \frac{b(j, m)}{4} + \frac{(2j - |m| + 1)^2 C}{4(1 - G(l))} \quad (49)$$

[with  $E_{oi} \rightarrow \infty$  for  $G(l) \geq 1$ ]. With a switch from Eq. (48) to (49) near  $G(l) = 0.5$  for the attractive channels, and with Eq. (49) for the repulsive channels, this provides a sufficiently accurate representation of the threshold energies  $E_{oi}(j, m, l)$ .

The situation changes to some extent if the exponent  $\gamma$  changes. We consider the case  $\gamma = 1/2$  which would correspond to a potential  $V(r) \propto r^{-4}$  with a shorter range and  $E_{cent} \propto r^{-2}$ . Fig. 5 shows the dependence of

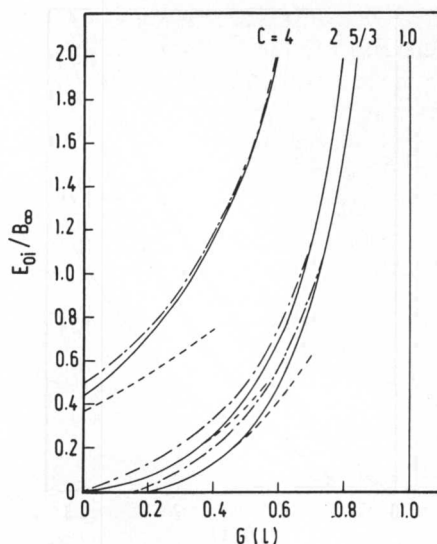


Fig. 5. As Fig. 4, but with  $\gamma = 1/2$  [i.e.  $V(r)$  (i.e.  $V(r) \propto r^{-4}$  and  $E_{\text{cent}} \propto r^{-2}$ ; —: accurate results, ---: oscillator expansion of Eq. (52), - - -: perturbation expansion of Eq. (51), - • - • -: PST result of Eq. (50)].

$E_{0i}$  on  $G(l)$  for the ( $j = 0, m = 0$ ) channel and the  $C$ -values 1 and 2. Phase space theory, for  $\gamma \leq 1$ , gives the result

$$\frac{E_o}{B_\infty} = \left( \frac{1}{\gamma} - 1 \right) (G(l)\gamma)^{\frac{1}{1-\gamma}}. \quad (50)$$

The  $C = 1$  curve now differs from PST; for  $C = 2$ , the threshold energies exceed the centrifugal barriers from PST even more. Like for  $\gamma = 1$ , the perturbation expansion applies well for small  $G(l)$ , whereas a switch to the oscillator expansion occurs in the range  $G(l) = 1 - 2$ . The general picture, therefore, does not differ basically from the PST threshold energies differ, the full range  $0 < G(l) < \infty$  is relevant, and the switch between the two limiting expressions occurs near  $G(l) \approx 1$  in contrast to  $G(l) \approx 0.5$  for  $\gamma = 1$ . For  $\gamma = 1/2$  and  $C = 2$ , the perturbation expansion gives

$$\frac{E_{0i}}{B_\infty} \approx j(j+1) + \frac{3F(j,m)}{2} \left[ \frac{G(l)}{F(j,m)\sqrt{2C}} \right]^{4/3}. \quad (51)$$

The oscillator expansion, for arbitrary  $C$ , gives

$$\frac{E_{0i}}{B_\infty} \approx \frac{b(j,m)}{4} + \frac{C}{4} \left[ 2j - |m| + 1 + \frac{G(l)}{\sqrt{C}} \right]^2. \quad (52)$$

Fig. 5 compares the results from PST, and from Eqs. (51) and (52) with the threshold energies derived from the tables in ref. [20].

The results of the present section have shown that  $E_{oi}$  cannot be expressed by a sum of centrifugal energies and  $E_{oi}$  ( $l = 0$ ). Instead, there is a considerable coupling. With increasing  $l$  and  $J$ , to an increasing extent  $E_{oi}$  approaches the range where the anharmonic oscillator expansion of the channel potential curves applies. Only when  $W(E, J = 0)$  near to the threshold energy is specifically relevant, or when thermal reaction rates are required for very low temperatures, i.e. selecting contributions from very small  $l$  only, the perturbation expansion dominates.

## 5. Coupling corrections

The simplified SACM treatment of refs. [6–8] has separated centrifugal effects and contributions from individual transition modes, see e.g. Eq. (10). The neglected couplings then were introduced a posteriori. Two types of corrections had to be applied:

(i) the violation of angular momentum couplings in the centrifugal effects was corrected via a comparison with the PST results;

(ii) the couplings between various transitional modes were corrected by defining [6–9] in a mostly intuitive way an effective Morse parameter  $\beta_{\text{eff}} > \beta$  which introduces more rigidity in an average way.

Obviously, there is some cancellation of errors in this procedure such that only small final adjustments of the ratio  $\alpha/\beta$  by comparison to a calibration experiment were required.

In the following, we investigate the coupling effects in a more detailed way. At first we consider the centrifugal effects. The representation of the channel threshold energies for  $j = 0$  in the previous section shows that, for channels with  $F(j, m) > 0$ , there appears to be no simple way of a separation or of an a posteriori correction.

However, when the channel threshold energies approach the oscillator results, a more general treatment can be made. As before, for  $\alpha/\beta = 0.5$ , the channel threshold energies  $E_{oi}$  here are given by the maxima of the approximate channel potential curves

$$\frac{V_i}{B_\infty} \approx -\frac{V_o}{CB_\infty} + (2j - |m| + 1) \sqrt{\frac{V_o}{B_\infty}} + \frac{b(j, m)}{4} + G(l) \left( \frac{V_o}{CB_\infty} \right)^\gamma. \quad (53)$$

Numerical results for  $E_{oi}(l, j, m)$  are easily obtained. Explicit results for  $\gamma = 1$  and  $\gamma = 1/2$  were given in Eqs. (49) and (52), respectively.

We now, for  $\gamma = 1$ , calculate the effects of the  $v \Leftrightarrow l$  coupling in activated complex partition functions and, hence, in thermal rate constants [1] (oscillator states with  $v = 2j - |m|$ , degeneracy  $v + 1$ , see above). Neglecting

the  $b(j, m)$ -term, for sufficiently large temperatures from Eq. (49) one obtains

$$Q^* = \sum_{v,l} \exp\left(-\frac{E_{oi}(v,l)}{kT}\right) \approx \int_0^\infty dv \int_0^1 dG v \exp\left(-\frac{v^2 CB_\infty}{4(1-G)kT}\right) \quad (54)$$

such that

$$Q^* \approx kT/B_\infty C. \quad (55)$$

If one would have separated the  $v$  and  $l$  contributions, with  $E_{oi}(v, l=0) = v^2 CB_\infty/4$  and  $E_{oi}(v=0, l) \approx 0$  the following result would have been obtained,

$$Q^* \approx \int_0^\infty dv \exp\left(-\frac{v^2 CB_\infty}{4kT}\right) \int_0^1 dG = \frac{\sqrt{\pi} 2kT}{CB_\infty}. \quad (56)$$

The coupling of the contributions from centrifugal effects and the two transition modes hence has introduced a "coupling factor" of  $f_c = 1/2\sqrt{\pi}$ . The overall rigidity factor [see Eq. (1)] from Eq. (55) follows as

$$f^{\text{rigid}} \approx 1/C. \quad (57)$$

The simplified SACM, with Eqs. (10), (42), and a global angular momentum correction [6] of  $4/\pi = 1/(\Gamma(3/2))^2$ , would have given  $f^{\text{rigid}} = 1/(1 + C/4)$ . In the charge-dipole case, with  $C = 2$ , this corresponds to  $f^{\text{rigid}} = 2/3$ . Applying the additional coupling factor  $f_c = 1/2\sqrt{\pi}$  would have given  $f^{\text{rigid}} = 0.19$ . Apparently, in this case the neglected couplings would have been markedly overcorrected. The oscillator result from Eq. (57) for  $C = 2$ , i.e.  $f^{\text{rigid}} \approx 0.5$ , is slightly above the accurate value [10]  $f^{\text{rigid}} = 0.42$  (approached at  $kT > B$ ). This is due to the fact that the oscillator approximation, for some attractive channels at  $l = 0$ , produces channel threshold energies which are below  $B_\infty j(j+1)$ .

For  $\gamma = 1/2$ , i.e. a  $V(r) \propto r^{-4}$  potential, the coupling effects appear somewhat different. From Eq. (52) one has

$$\begin{aligned} Q^* &\approx \int_0^\infty dv \int_0^\infty dG v \exp\left[-\frac{CB_\infty}{4kT}\left(v + \frac{G}{\sqrt{C}}\right)^2\right] \\ &= \left(\frac{4kT}{CB_\infty}\right)^{3/2} \sqrt{C} \int_0^\infty dx \int_0^\infty dy x \exp[-(x+y)^2] = \left(\frac{4kT}{CB_\infty}\right)^{3/2} \frac{\sqrt{\pi C}}{8}. \end{aligned} \quad (58)$$

Separation would have given

$$Q^* \approx \int_0^\infty v \exp\left(-\frac{CB_\infty}{4kT}v^2\right) dv \int_0^\infty \exp\left(-\frac{B_\infty}{4kT}G^2\right) dG = \left(\frac{4kT}{CB_\infty}\right)^{3/2} \frac{\sqrt{\pi C}}{4} \quad (59)$$

such that a coupling factor of  $f_c = 1/2$  would have been required. With the PST result for  $\gamma = 1/2$  from Eq. (50) one obtains

$$Q^{*PST} = \frac{kT}{B_\infty} \int_0^\infty dG \exp\left(-\frac{G^2 B_\infty}{4kT}\right) = \sqrt{\pi} \left(\frac{kT}{B_\infty}\right)^{3/2} \quad (60)$$

such that, with Eq. (58), one has like in Eq. (57)

$$f^{\text{rigid}} \approx 1/C. \quad (61)$$

For  $C = 2$ , this gives  $f^{\text{rigid}} = 0.5$ , compared to the simplified SACM result  $f^{\text{rigid}} = 1/(1 + C/4) = 0.67$ . Applying a coupling factor of  $f_c = 1/2$  here would lead to a better agreement. It appears that the  $\gamma = 1$  case with its particularly long range of the potential  $V(r)$  behaves unusually because of the large contribution from attractive channels. Then, the simplified SACM model with an additional coupling factor does not work too well. For the  $\gamma = 1/2$  case, which would be more typical for potentials with a shorter range, the model apparently works better.

On the level of the oscillator expansion we now consider the coupling of  $b$  transition modes. Eq. (53), for  $b$  non-degenerate transition modes and omitting the  $b/4$ - and the centrifugal terms, is replaced by

$$V_i \approx -|V| + \sum_{n=1}^b (v_n + 1/2) \sqrt{C_n |V| B_{\infty n}} \quad (62)$$

where  $C_n$  are the relevant factors in Eq. (36) and  $B_{\infty n}$  are the rotational constants for the correlating fragments of the  $b$  individual transition modes. The channel threshold energies corresponding to Eq. (62) are now calculated to be

$$E_{oi} \approx \left[ \sum_{n=1}^b (v_n + 1/2) \sqrt{C_n B_{\infty n}} \right]^2 4. \quad (63)$$

The corresponding activated complex partition function follows as

$$Q^* \approx \int_0^\infty \dots \int_0^\infty dv_1 \dots dv_b \exp \left[ -\frac{\left( \sum_{n=1}^b v_n \sqrt{C_n B_{\infty n}} \right)^2}{4kT} \right] \quad (64)$$

such that, after evaluation of the integrals,

$$Q^* \approx c(b) \prod_{n=1}^b \sqrt{\frac{4kT}{C_n B_{\infty n}}} \quad (65)$$

with  $c(2) = 1/2$ ,  $c(3) = \sqrt{\pi}/8$ ,  $c(4) = 1/12$ ,  $c(5) = \sqrt{\pi}/64$ , and  $c(6) = 1/120$ . Division by the corresponding fragment rotational partition functions gives the total rigidity factors which are summarized in Table 1. A compari-

**Table 1.** Approximate rigidity factors  $f^{\text{rigid}}$  including coupling factors  $f_c$  [see Eqs. (16), (65), and (66)].

| Fragments                   | $b$ | $f^{\text{rigid}}$            | $f_c$      |
|-----------------------------|-----|-------------------------------|------------|
| Linear + atom               | 2   | $C^{-1}$                      | 1          |
| Nonlinear + atom            | 3   | $C^{-3/2}$                    | 1          |
| Linear(1) + linear(2)       | 4   | $3C_1^{-1}C_2^{-1}/4$         | 3/4        |
| Nonlinear(1) + linear (2)   | 5   | $C_1^{-3/2}C_2^{-1}/4$        | 1/4        |
| Nonlinear(1) + nonlinear(2) | 6   | $4C_1^{-3/2}C_2^{-3/2}/15\pi$ | 4/15 $\pi$ |

son with Eqs. (58) and (59) indicates that, for a  $V(r) \propto r^{-4}$  radial potential, the including of centrifugal effects in addition would require a centrifugal coupling factor of 1/2 which is independent of the value of  $b$ . This factor is included in the rigidity factors of Table 1.

## 6. Conclusions

In the present work, the statistical adiabatic channel model of ref. [2] and its simplified versions from refs. [6–8] have been compared with a treatment using explicit channel potential curves. The channel eigenvalue interpolation parameters  $\alpha$  and  $\beta$ , such as used in refs. [2, 6–8], are reinterpreted in terms of properties characterizing the anisotropy of the potential. In particular, the ratio  $2\alpha/\beta$  is identified with the exponent of a relation between the anisotropy barriers  $V_o(r)$  and the radial potential  $V(r)$  via the equation

$$\frac{V_o(r)}{B_\infty} \approx C \left( \frac{|V(r)|}{B_\infty} \right)^{2\alpha/\beta} \quad (36)$$

The parameter  $C$  describes the height of the anisotropy barriers.  $C = 2$  corresponds to a  $\cos \theta$ -angular potential and  $C = 1$  to an anisotropic potential with barriers just reaching up to the  $E = 0$  plane. The reality will often be in between this [11, 12, 18], i.e.  $1 < C < 2$ .

A number of experimental observables sensitively depend on the potential properties such as the parameters  $\alpha/\beta$ ,  $\gamma$ , and  $C$ . E.g., specific rate constants  $k(E, J = 0)$  for dissociation near to the threshold energy  $E = 0$  may follow the predictions from phase space theory, before they fall below with increasing energy, if the parameter  $C$  is smaller than 2. Such behaviour has been discovered recently by experiment [31]. For  $C > 2$ , instead an SACM-modified PST [9] quite generally applies, with the rotational constant  $B_\infty$  replaced by  $B_{\text{eff}} > B_\infty$  independent of the energy. The ratio  $B_{\text{eff}}/B_\infty$ , for  $2\alpha/\beta = 1$ , is related to the parameter  $C$  either, for small  $J$ , via

$$B_{\text{eff}}/B_\infty \approx \text{Max} \{1, C - 1/2\} \quad (40)$$

or, for large  $J$ , via

$$B_{\text{eff}}/B_{\infty} \approx 1 + C/4. \quad (42)$$

Rigidity factors  $f^{\text{rigid}} = k/k^{\text{PST}}$  [see Eq. (1)] in thermal dissociation and recombination reactions are given by the product of  $B_{\infty}/B_{\text{eff}}$  factors and a coupling factor  $f_c$  (see Table 1), i.e. with Eq. (42) by

$$f^{\text{rigid}} \approx f_c \prod_{n=1}^b \left( \frac{1}{1 + C_n/4} \right)^{1/2} \quad (66)$$

or the equivalent expression using Eq. (40).  $b$  includes all transition modes which are correlating reactant oscillator (quantum  $\varepsilon_{\text{en}}$  at  $r = r_e$ ) and product rotor coordinates (rotational constant  $B_{\infty n}$  at  $r \rightarrow \infty$ ). Coupling factors  $f_c$  are given in Table 1 for various values of  $b$  between 2 and 6 [using a radial potential  $V(r) \propto r^{-4}$  which has a shorter range than the  $r^{-2}$ -centrifugal potential].

When accurate potential surfaces are available, one may try to identify  $C$ ,  $\gamma$ , and  $2\alpha/\beta$  for the relevant range of interfragment distances  $r$ . If this is not the case,  $C$  may be estimated from

$$C \approx \varepsilon_e^2/2DB_{\infty} \quad (67)$$

[for  $2\alpha/\beta = 1$ , see Eq. (22), Morse dissociation energy  $D$ ] or from the leading long-range potential expressed in the form of Eq. (36) (if capture reactions without a stable complex are considered).

Apart from identifying  $B_{\text{eff}}$  via Eqs. (42) and (64), the coupling of transition modes among each other and with the centrifugal contributions is of crucial importance. In the simplified SACM of refs. [6–9] this was done by a global angular momentum coupling correction and by the more intuitive use of an effective Morse parameter  $\beta_{\text{eff}} > \beta$ . The present work puts this correction procedure on a more serious basis.

In the quick comparison with experimental results on thermal rate coefficients, artifacts of the simplified SACM treatment in ref. [1] were compensated by minor adjustments of the ratio  $\alpha/\beta$ . In Appendix A, for some selected reactions from ref. [1], we compare experimental rigidity factors with the present model using Eqs. (66), (67), and Table 1 based on the assumption of  $\alpha/\beta = 0.5$ . Taking into account that the experiments were interpreted with centrifugal partition functions from a Morse radial potential, which may result in errors up to a factor of 2, the agreement with the present simple method appears satisfactory.

It should finally be emphasized that the few illustrative but explicit calculations of this work cannot replace a detailed treatment of individual reaction systems. Their individual radial potentials  $V(r)$ , anisotropy barriers  $V_o(r)$  and angular dependences generally will have a more complex form

than used here. Such numerical calculations are underway for the [11]  $\text{H}_2\text{O}_2$  and the [12]  $\text{HO}_2$  systems.

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## Appendix A

### Simplified determination of rigidity factors $f^{\text{rigid}}$ for thermal rate coefficients

The treatment of this work has led to very simple expressions for the rigidity factors  $f^{\text{rigid}}$  in the thermal rate coefficients of Eq. (1) for simple unimolecular bond fission-radical recombination reactions. We illustrate the success of these expressions by a comparison with experimental results for a number of selected reactions such as considered in ref. [1].

In the simplest procedure one inspects the frequencies of the oscillators of the reactant A and of the products B + C. One omits all oscillators of A which are "conserved", i.e. which (with some frequency changes) become fragment oscillators. One also omits one oscillator for the reaction coordinate. In addition, one omits all "disappearing" free or hindered internal rotors (or low frequency torsional-type reactant oscillators) and the K-rotor (for a non-linear) reactant. One then calculates the geometrical mean  $\langle \epsilon_e \rangle$  of the quanta of the remaining  $b^*$  conserved oscillators of the reactant A, and one calculates the geometrical mean  $\langle B_\infty \rangle$  of all rotational constants of the fragments B + C. One finally identifies D with the reaction enthalpy  $\Delta H_0^\circ$ . Then an average potential parameter  $\langle C \rangle$  by analogy to Eq. (41) is obtained via

$$\langle C \rangle = \frac{(\langle \epsilon_e \rangle / \Delta H_0^\circ)^2}{2(\langle B_\infty \rangle / \Delta H_0^\circ)} \quad (\text{A1})$$

The rigidity factors  $f^{\text{rigid}}$  then follow by analogy to Table 1 as

$$f^{\text{rigid}} \approx 1/\langle C \rangle \quad \text{for } b^* = 2 \quad (\text{A2})$$

$$f^{\text{rigid}} \approx 1/(\langle C \rangle^{3/2}) \quad \text{for } b^* = 3 \quad (\text{A3})$$

$$f^{\text{rigid}} \approx 3/(4\langle C \rangle) \quad \text{for } b^* = 4. \quad (\text{A4})$$

The largest possible  $b^*$ -value (for a nonlinear reactant) is 4. The case  $b^* = 1$  of a nonlinear reactant A dissociating into an atom B and a linear fragment C has not yet been treated. In this case, the K-rotor of the reactant leads over to rotational states of the linear fragment. The corresponding rigidity factor will be between the square root of Eq. (A2) and Eq. (2). Choosing the geometrical mean, we employ

$$f^{\text{rigid}} \approx 1/(\langle C \rangle^{3/4}) \quad \text{for } b^* = 1. \quad (\text{A5})$$

At the next, slightly more sophisticated level, one may include all disappearing oscillators and internal free or hindered rotors (including the K-rotor) of the reactant, calculate for each individual of these coordinates a  $C_i$ -value via

$$C_i \approx 1 + \epsilon_{ei}^2 / 8 \Delta H_0^\circ B_{\infty i} \quad (\text{A6})$$

**Table 2.** Calculated and experimental rigidity factors for thermal high-pressure recombination reactions (see ref. [1]).

| Reaction                                                          | $\Delta H_0^\circ$<br>cm <sup>-1</sup> | $\epsilon_e/$ | $\langle B_\infty \rangle/$ | $b^*$ | $\langle C \rangle$ | $f_{\text{calc.}}^{\text{rigid}}$ | $f_{\text{exp.}}^{\text{rigid}}$ |
|-------------------------------------------------------------------|----------------------------------------|---------------|-----------------------------|-------|---------------------|-----------------------------------|----------------------------------|
| H + O <sub>2</sub> → HO <sub>2</sub>                              | 16800                                  | 1101          | 1.445                       | 1     | 25.0                | 0.089                             | 0.10                             |
| HO + O → HO <sub>2</sub>                                          | 22600                                  | 1101          | 18.871                      | 1     | 1.42                | 0.77                              | 0.64                             |
| • H + OH → H <sub>2</sub> O                                       | 41300                                  | 1595          | 18.871                      | 1     | 1.63                | 0.69                              | 0.85                             |
| O + O <sub>2</sub> → O <sub>3</sub>                               | 8460                                   | 705           | 1.445                       | 1     | 20.3                | 0.10                              | 0.17                             |
| O + NO → NO <sub>2</sub>                                          | 25150                                  | 757           | 1.704                       | 1     | 6.69                | 0.24                              | 0.74 <sup>a</sup>                |
| Cl + NO → ClNO                                                    | 13010                                  | 595           | 1.704                       | 1     | 7.98                | 0.21                              | 0.89                             |
| I + NO → INO                                                      | 6050                                   | 200           | 1.704                       | 1     | 1.94                | 0.61                              | 0.57                             |
| HO + HO → H <sub>2</sub> O <sub>2</sub>                           | 17350                                  | 1387          | 18.871                      | 2     | 2.68                | 0.37                              | 0.38                             |
|                                                                   |                                        | 1266          |                             |       |                     |                                   |                                  |
| H + CH <sub>3</sub> → CH <sub>4</sub>                             | 36200                                  | 1533          | 7.602                       | 2     | 4.27                | 0.23                              | 0.19                             |
|                                                                   |                                        | 1533          |                             |       |                     |                                   |                                  |
| NO <sub>2</sub> + NO <sub>3</sub> → N <sub>2</sub> O <sub>5</sub> | 7690                                   | 577           | 0.624                       | 3     | 18.0                | 0.013                             | 0.0062                           |
|                                                                   |                                        | 353           |                             |       |                     |                                   |                                  |
|                                                                   |                                        | 353           |                             |       |                     |                                   |                                  |
| NO <sub>2</sub> + NO <sub>2</sub> → N <sub>2</sub> O <sub>4</sub> | 4440                                   | 675           | 1.125                       | 4     | 23.2                | 0.0014                            | 0.0020                           |
|                                                                   |                                        | 480           |                             |       |                     |                                   |                                  |
|                                                                   |                                        | 430           |                             |       |                     |                                   |                                  |
|                                                                   |                                        | 385           |                             |       |                     |                                   |                                  |
| CH <sub>3</sub> + CH <sub>3</sub> → C <sub>2</sub> H <sub>6</sub> | 30850                                  | 1206          | 7.602                       | 4     | 2.11                | 0.17                              | 0.15                             |
|                                                                   |                                        | 1206          |                             |       |                     |                                   |                                  |
|                                                                   |                                        | 821           |                             |       |                     |                                   |                                  |
|                                                                   |                                        | 821           |                             |       |                     |                                   |                                  |

<sup>a</sup>  $f_{\text{exp.}}^{\text{rigid}}$  includes a contribution from electronically excited NO<sub>2</sub> which is not included in  $f_{\text{calc.}}^{\text{rigid}}$ .

and put these  $C_i$ -values into the expression

$$f^{\text{rigid}} \approx d(b) \prod_{i=1}^b C_i^{-1/2}$$

(A7)

with  $d(2) = d(3) = 1$ ,  $d(4) = 3/4$ ,  $d(5) = 1/4$ , and  $d(6) = 4/15\pi$  by analogy with Table 1. In view of the uncertainties in the centrifugal partition function entering  $k^{\text{PST}}$  we do not apply this approach but use the simpler result of Eqs. (A2)–(A5) in the following Table 2.

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